

Manual of HDAC8 Assistant open-source software

HDAC8 Assistant is a web application for the generation and evaluation of chemical compounds as potential HDAC8 inhibitors. The software combines *de novo* molecule generation (SMILES-RNN with reinforcement learning), QSAR-based activity prediction, drug-likeness evaluation (Muegge rules), and structural alert screening (Brenk filters and PAINS).

Appearance of the main window

The main page displays five capability panels: Molecule generation, Machine learning (QSAR), OECD guidelines (applicability domain), Muegge's rules, and Structural Alerts. A download button provides this manual.

HDAC8 Assistant

The app enables the generation and evaluation of chemicals as HDAC8 inhibitors.

Molecule generation	Machine learning	OECD rules	Muegge's rules	Structural Alerts
				
This application generates candidate HDAC8 inhibitors using a SMILES-RNN model pretrained on ChEMBL drug-like compounds. The model is optimized via reinforcement learning (REINVENT) to maximize predicted pIC50 values and ensure applicability domain compliance. An ECFP4-based CatBoost QSAR model serves as the core of the RL reward function. Generated structures undergo deduplication, validity checks, and optional ring-system filtering; they are then ranked by predicted activity and annotated with synthetic accessibility scores (SAScore) and Muegge drug-likeness criteria.	This application makes predictions based on Quantitative Structure-Activity Relationship (QSAR) models built on curated datasets generated from scientific articles. The models were developed using open-source chemical descriptors based on ECFP4, along with the gradient boosting method.	We follow the best practices for model development and validation recommended by guidelines of the Organization for Economic Cooperation and Development (OECD). The applicability domain (AD) of the models was calculated as $D_{cutoff} = (D) + Z_s$, where Z_s is a similarity threshold parameter defined by a user (0.5 in this study) and $\langle D \rangle$ and $\langle s \rangle$ are the average and standard deviation, respectively, of all Euclidean distances in the multidimensional descriptor space between each compound and its nearest neighbors for all compounds in the training set.	Estimating the drug-likeness of a compound is an important factor in drug development. Muegge's drug-likeness rules were introduced to estimate the potential of a compound to be a drug. Our drug-likeness radar is displayed for a quick assessment of the compliance of the tested compound with the Muegge rules. The application also provides structural analysis for identifying preferred or undesirable molecular fragments.	Brenk filters which consists in a list of 105 fragments to be putatively toxic, chemically reactive, metabolically unstable or to bear properties responsible for poor pharmacokinetics. PAINS are molecules containing substructures showing potent response in assays irrespective of the protein target. Such fragments, yielding false positive biological output.

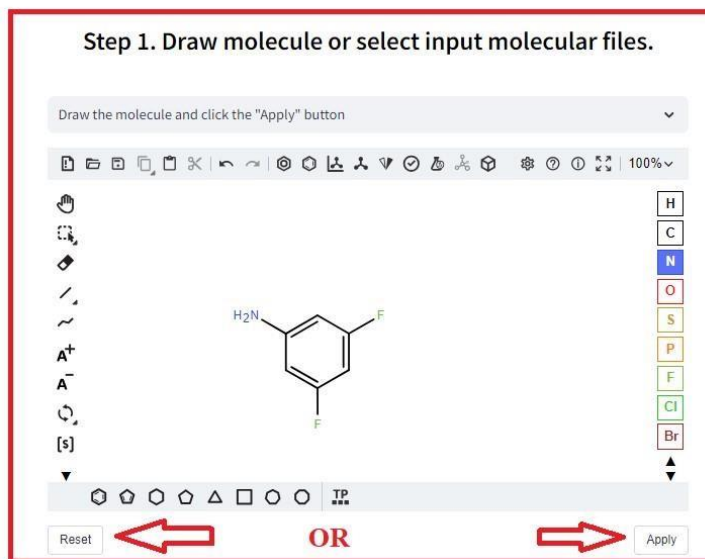
[Click to download brief manual](#)

Step 1. Draw molecule or select input molecular files

If you want to draw the structure of a chemical compound, use the built-in chemical editor Ketcher (<https://github.com/epam/ketcher>). After creating the structure, use two buttons: (1) "Reset" - delete the structure to create a new one; (2) "Apply" - transfer the structure for further analysis or generation in Step 2.

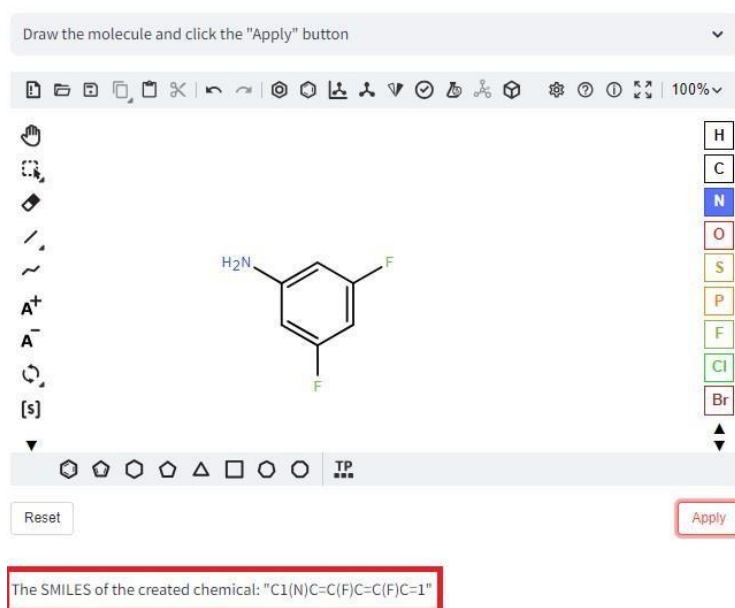
Step 1. Draw molecule or select input molecular files.

Draw the molecule and click the "Apply" button



Reset **OR** Apply

After clicking the "Apply" button, make sure that the structure has been created. If successful, its SMILES will be displayed under the chemical editor.

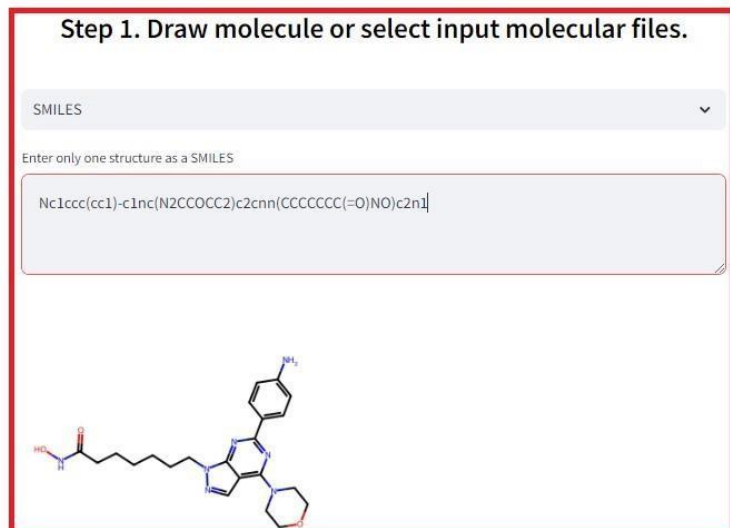


If you choose SMILES, paste the SMILES representation of the desired chemical structure and press Ctrl+Enter. If the entered structure is correct, the application will generate a 2D image of the compound.

DON'T FORGET TO CLICK THE "APPLY" BUTTON when using the Ketcher editor.

If the entered structure is incorrect, the application reports an error.

If Ketcher export fails aromatic bond assignment (kekulization), switch to the SMILES input method and paste the same structure as text.



Step 1. Draw molecule or select input molecular files.

SMILES

Enter only one structure as a SMILES

ON(C)(=O)c1ccc(CCCCc2ccccc2)cc1

ArgumentError: Python argument types in rdkit.Chem.rdmolfiles.MolToSmiles(NoneType) did not match C++ signature: MolToSmiles(class RDKit::ROMol mol, bool isomericSmiles=True, bool kekulizeSmiles=False, int rootedAtAtom=-1, bool canonical=True, bool allBondsExplicit=False, bool allHsExplicit=False, bool doRandom=False) MolToSmiles(class RDKit::ROMol mol, struct RDKit::SmilesWriteParams params)

Traceback:

```
File "C:\ProgramData\anaconda3\envs\my-rdkit-env\lib\site-packages\streamlit\r
exec(code, module.__dict__)
File "C:\Users\user\My_apps\Toxicity_Estimator\Toxicity_Estimator_app.py", lin
canon_smi = Chem.MolToSmiles(Chem.MolFromSmiles(compound_smiles),isomericS
```

If you choose a file *.sdf or *.csv containing multiple chemical structures, click the "Browse files" button and upload the file. For *.csv files, the file must contain a column named "SMILES".

Select input molecular files

*CSV file containing SMILES

The file should contain a column with the name "SMILES"

Drag and drop file here
Limit 200MB per file

Browse files

saved_example_2.csv 0.6KB

CHEMICAL STRUCTURE VALIDATION AND STANDARDIZATION:

Original data: 11 molecules

Failed data: 0 molecules

Kept data: 11 molecules

Run predictions!

If incorrect structures are detected in an *.sdf or *.csv file, the corresponding information appears in the section "CHEMICAL STRUCTURE VALIDATION AND STANDARDIZATION".

MDL multiple SD file (*.sdf) ▼

Choose a SDF file

 Drag and drop file here
 Limit 200MB per file

Browse files

 211_prop.sdf 0.7MB
 ✕

CHEMICAL STRUCTURE VALIDATION AND STANDARDIZATION:

Original data: 211 molecules

Failed data: 9 molecules

No.	No. failed molecule in original set	SMILES of wrong structure:
1	80	<chem>COC1=C/C2=C(OC3=CC=C(NC(=O)C4=C(O)C=CN(C5=CC=CC=C5)=N4)C=C3</chem>
2	88	<chem>O=C/C=C/C1=CC=C(CCN(C(=O)C2=CC(C/C3=N/N=C(O)C4=CC=CC=C43)=CC</chem>
3	90	<chem>COC1=C\C(OC)=C2\C(=O)=NC(C3=CC(C)=C(OCCCCC(=O)NO)C(C)=C3)=N\C2</chem>
4	92	<chem>COC1=C/C2=C(OC3=CC=C(NC(=O)C4=C(O)C(C)=CN(C5=CC=CC=C5)=N4)C=</chem>
5	93	<chem>CCC1=CN(C2=CC=CC=C2)=NC(C(=O)NC2=CC=C(O/C3=C/C=N\C4=CC(OCCCC</chem>
6	153	<chem>O=C(NO)C1=CC=C(CN2CCN(C(=O)C3=CC(C/C4=N/N=C(O)C5=CC=CC=C54):</chem>
7	158	<chem>O=C(CC1=CC=C(CN2CCN(C(=O)C3=CC(C/C4=N/N=C(O)C5=CC=CC=C54)=C</chem>
8	193	<chem>O=C/C=C/C1=CC=C(CN2CCN(C(=O)C3=CC(C/C4=N/N=C(O)C5=CC=CC=C54</chem>
9	203	<chem>COC1=C\C(OC)=C2\C(=O)=NC(C3=CC(C)=C(OCC4=CC=C/C=C/C(=O)NO)C=C</chem>

Kept data: 202 molecules

Step 2. Select analysis mode

In Step 2, choose one of the following modes from the drop-down menu:

- HDAC8 - predict HDAC8 inhibitor activity (pIC50) for the input structure(s);
- Molecule generation- .

Step 2a. HDAC8 activity prediction

Select "HDAC8" and click the "Run predictions!" button. The QSAR model uses ECFP4 molecular fingerprints and a CatBoost gradient boosting regressor trained on curated HDAC8 activity data from scientific literature and ChEMBL.

Step 2. Select prediction of HDAC8 inhibitor activity or substructural search for preferred or undesirable fragments



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Click on the "Run predictions!" button for prediction. **Step**

Prediction results (HDAC8 mode)

The form of presentation depends on the input format. For a single molecule (Ketcher or SMILES), results are displayed on screen. For *.sdf or *.csv files, results for valid structures are shown in a downloadable table.

For example, when selecting SMILES, the results will be displayed for a single molecule. When displaying the results on the screen, it is taken into account whether there are experimental values of activity for the studied compound.

The final table 'Prediction results' contains the following columns:

- 1) **SMILES** - the chemical structure is displayed in the SMILES notation
- 2) **Predicted value pIC₅₀** - the predicted value of the activity to inhibit the HDAC8 enzyme, expressed in pIC₅₀, where pIC₅₀ is the negative logarithm of IC₅₀ in molar concentration. If experimental data is available in the ChEMBL database, the label "see experimental value" is displayed in this cell.
- 3) **Applicability domain_HDAC8** - compliance of the chemical compound with Applicability domain. If experimental data is available in the ChEMBL database, the label "-" is displayed in this cell.
- 4) **Experimental value, pIC₅₀** - experimental data presented in the ChEMBL database. Where pIC₅₀ is the negative logarithm of IC₅₀ in molar concentration. If there is more than one value in the database, then the average value is given in the cell.
- 5) **STD** - the cell indicates the standard deviation of the experimental activity values presented in the ChEMBL database.
- 6) **chemble_ID** - identifier from the ChEMBL chemical database of molecule

If you choose a file *.sdf or *.csv, the prediction results for correct chemical structures are displayed in a table that can be downloaded.

Original data: 11 molecules

Failed data: 0 molecules

Kept data: 11 molecules

Run predictions!

	SMILES	Predicted value, pIC50	Ap
2	<chem>CCOC1cc(NC(=O)CCCCC(=O)Nc2cc(-c3cccc3)ccc2N)cc2c1OCC([N+](=O)[O-])=C2</chem>	see experimental value	-
3	<chem>[N-]=[N+]=NCCC(=O)Nc1ccc(C(=O)Nc2cc(N=[N+]=[N-])ccc2N)cc1</chem>	see experimental value	-
4	<chem>[N-]=[N+]=Nc1cccc(COC(=O)Nc2ccc(C(=O)Nc3cc(-c4cccc4)ccc3N)cc2)c1</chem>	see experimental value	-
5	<chem>[N-]=[N+]=NCc1cc(N=[N+]=[N-])cc(C(=O)Nc2cc(-c3cccs3)ccc2N)c1</chem>	see experimental value	-
6	<chem>C#CCOc1ccc(C=NNC(=O)c2ccc(C(=O)NO)c2)cc1</chem>	see experimental value	-
7	<chem>O=C(NO)c1cnc(NC2(c3cccc3)CCC(F)(F)CC2)nc1</chem>	see experimental value	-
8	<chem>CN1CCc2c(c3cccc3n2Cc2ccc(C(=O)NO)cc2)C1</chem>	see experimental value	-
9	<chem>CCCNNC(=O)c1ccc(CNC(=O)C(Cc2c[nH]c3cccc23)NC(=O)c2ccc(OC)cc2)cc1</chem>	see experimental value	-
10	<chem>COc1ccc(C(=O)c2ccc3c(ccn3Cc3ccc(C=CC(=O)NO)cc3)c2)cc1</chem>	see experimental value	-
11	<chem>CC(C=CC(=O)NO)=CC(C)C(=O)c1ccc(N(C)C)cc1</chem>	see experimental value	-

Download results of prediction as CSV

Step 2b. Molecule generation (SMILES-RNN)

Select "Molecule generation (SMILES-RNN)" in Step 2. This mode generates new candidate HDAC8 inhibitors using a SMILES-RNN model pretrained on drug-like compounds from ChEMBL. The model is optimized via reinforcement learning (REINVENT algorithm) to maximize predicted pIC50 values while ensuring applicability-domain compliance. An ECFP4-based CatBoost QSAR model serves as the core of the RL reward function.

Step 2. Select prediction of HDAC8 inhibitor activity, molecular generation, or substructural analysis

Molecule generation (SMILES-RNN) ▼

Run SMILES-RNN RL locally. Final results include only molecules inside AD, and exclude compounds that match the experimental pIC50 dataset (HDAC8_exp_data_inchi.csv).

Mode: Run SMILES-RNN RL now (GPU/CPU auto-selection).

Maximum molecules to return

500 - +

RL n_steps

250 - +

RL batch_size

128 - +

Run generation!

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For the Molecule Generation (SMILES-RNN) stage, the user initiates de novo generation of novel potential HDAC8 inhibitors using a SMILES-RNN model. The generation process is performed locally on the user's computer. If a GPU is available, the application automatically utilizes it; otherwise, all computations are executed on the CPU.

Before starting the generation process, the user specifies three key parameters.

Maximum Molecules to Return

This parameter determines the maximum number of molecules displayed in the final results table after generation and filtering. Importantly, it does not correspond to the total number of molecules produced by the model during training; rather, it defines the number of final candidates that successfully pass the post-processing pipeline. The application retains only chemically valid structures, removes duplicates, verifies whether molecules fall within the applicability domain of the QSAR model, excludes compounds already present in the experimental HDAC8 dataset, and subsequently ranks the remaining molecules according to their predicted pIC50 values. For example, a value of 500 indicates that no more than 500 top-ranked filtered candidates will be returned to the user.

RL n_steps

This parameter specifies the number of reinforcement learning iterations. During each iteration, the SMILES-RNN model generates a batch of SMILES structures, which are subsequently evaluated using a reward function. The reward incorporates both the predicted HDAC8 inhibitory activity and the molecule's membership within the applicability domain of the QSAR model. Higher values of RL n_steps allow the model to optimize longer toward more promising chemical structures but also increase computational time. The default value of 250 represents a practical balance between generation quality and execution speed.

RL batch_size

This parameter defines the number of molecules generated and evaluated during a single reinforcement learning iteration. For example, with batch_size = 128, the model generates 128 SMILES candidates at each step, which are then assessed by the QSAR model and the applicability-domain filter. Increasing the batch size may improve the diversity of molecules explored during each iteration but requires additional system memory and GPU memory. The default value of 128 is suitable for most generation tasks.

After configuring the parameters, the user clicks **Run Generation!**. The application launches the SMILES-RNN model, performs reinforcement learning optimization, and then processes the generated compounds through a post-processing workflow. This workflow removes invalid SMILES strings, canonicalizes molecular structures, eliminates duplicates, retains only molecules within the applicability domain, ranks candidates according to their

predicted_pIC50 values, and calculates additional molecular properties, including the Synthetic Accessibility Score (SAScore) and compliance with the Muegge drug-likeness criteria.

The final generation results are presented in a downloadable table containing the prioritized candidate HDAC8 inhibitors.

Generation results:

Initial generated	Valid unique	Inside AD (returned)	Excluded (exp. dataset)
12800	2300	100	8

Execution mode/device: gpu

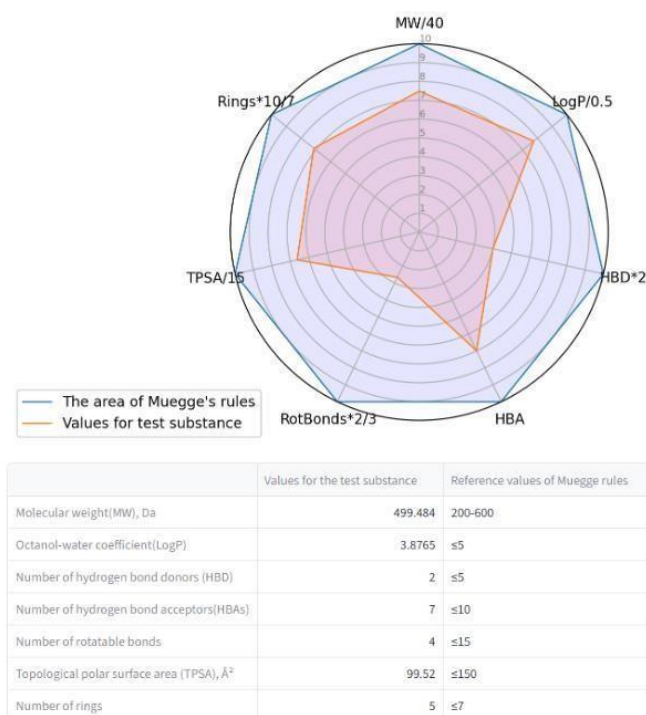
No.	step	generated_smiles	canonical_smiles	predicted_pIC50	SAScore	in_AD	Muegge rules
1	18	<chem>O=C(NO)c1cccc(NS(=O)(=O)c2ccccc2)c1</chem>	<chem>O=C(NO)c1cccc(NS(=O)(=O)c2ccccc2)c1</chem>	7.0578	1.728	☒	yes
2	15	<chem>O=C(NO)c1ccc(NS(=O)(=O)c2ccc(-c3ccccc3)cc2)cc1</chem>	<chem>O=C(NO)c1ccc(NS(=O)(=O)c2ccc(-c3ccccc3)cc2)cc1</chem>	6.8555	1.764	☒	yes
3	13	<chem>O=C(NO)c1cccc(NS(=O)(=O)c2ccc(Cl)cc2)c1</chem>	<chem>O=C(NO)c1cccc(NS(=O)(=O)c2ccc(Cl)cc2)c1</chem>	6.7933	1.792	☒	yes
4	15	<chem>O=C(NO)c1ccc(NC(=O)c2ccc3ccccc3c2)cc1</chem>	<chem>O=C(NO)c1ccc(NC(=O)c2ccc3ccccc3c2)cc1</chem>	6.7741	1.669	☒	yes
5	20	<chem>CN(C)c1ccc(C(=O)Nc2ccc(C(=O)NO)cc2)cc1</chem>	<chem>CN(C)c1ccc(C(=O)Nc2ccc(C(=O)NO)cc2)cc1</chem>	6.6859	1.732	☒	yes
6	19	<chem>O=C(NO)c1ccc(CO)cc1</chem>	<chem>O=C(NO)c1ccc(CO)cc1</chem>	6.6487	1.689	☒	no, 1 violator
7	19	<chem>Cc1cc(C(=O)NO)cccc1=O</chem>	<chem>Cc1cc(C(=O)NO)cccc1=O</chem>	6.6459	2.077	☒	no, 1 violator
8	18	<chem>O=C(NO)c1ccc(C(=O)Nc2ccc(Cl)cc2)cc1</chem>	<chem>O=C(NO)c1ccc(C(=O)Nc2ccc(Cl)cc2)cc1</chem>	6.5957	1.552	☒	yes
9	17	<chem>O=C(NO)c1cccc(-c2ccc(CO)cc2)c1</chem>	<chem>O=C(NO)c1cccc(-c2ccc(CO)cc2)c1</chem>	6.579	1.731	☒	yes
10	18	<chem>OC1ccc(NC(=O)c2ccc(C(=O)NO)cc2)cc1Cl</chem>	<chem>OC1ccc(NC(=O)c2ccc(C(=O)NO)cc2)cc1Cl</chem>	6.5103	1.697	☒	yes

Download generated molecules as CSV

Step 3. Muegge rules and structural alerts

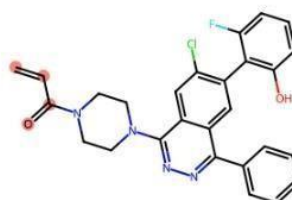
The HDAC8 Assistant application can be used to evaluate the compliance of the compound to be studied with Muegge's rules, and the result is displayed both in tabular numerical form and graphically, in the form of the so-called Bioavailability Radar, which can be used to visually evaluate the compliance of the calculated values of lipophilicity (log P), molecular weight (MW), and the number of hydrogen bond donors (HBD) and hydrogen bond acceptors (HBA) and other parameters with the limits determined by Muegge's rules.

The Drug-likeness Radar: compliance with Muegge's rules



In addition, the HDAC8 Assistant web application integrates detection of such widely used in medicinal chemistry structural alerts as Brenk filters and PAINS when analyzing single structures. Brenk filters which consists in a list of 105 fragments identified by Brenk et al. [Brenk, R. et al. Lessons learnt from assembling screening libraries for drug discovery for neglected diseases. *ChemMedChem* 3, 435– 444 (2008).] to be putatively toxic, chemically reactive, metabolically unstable or to bear properties responsible for poor pharmacokinetics. PAINS (for pan assay interference compounds, a.k.a. frequent hitters or promiscuous compounds) are molecules containing substructures showing potent response in assays irrespective of the protein target. Such fragments, yielding false positive biological output, have been identified by Baell et al. [Baell, J. B. & Holloway, G. A. New substructure filters for removal of pan assay interference compounds (PAINS) from screening libraries and for their exclusion in bioassays. *J. Med. Chem.* 53, 2719–2740 (2010). HDAC8_Predictor returns warnings if such moieties are found in the molecule under evaluation.

The found Brenk filter: Michael-acceptor



The PAINS are not found in the molecule.

Software requirements for molecule generation

Molecule generation requires the Python packages smiles-rnn, molscore, and PyTorch (see requirements.txt). Optional packages useful-rdkit-utils (ring-system filtering) and sascorer (SAScore calculation) enhance post-processing when available.